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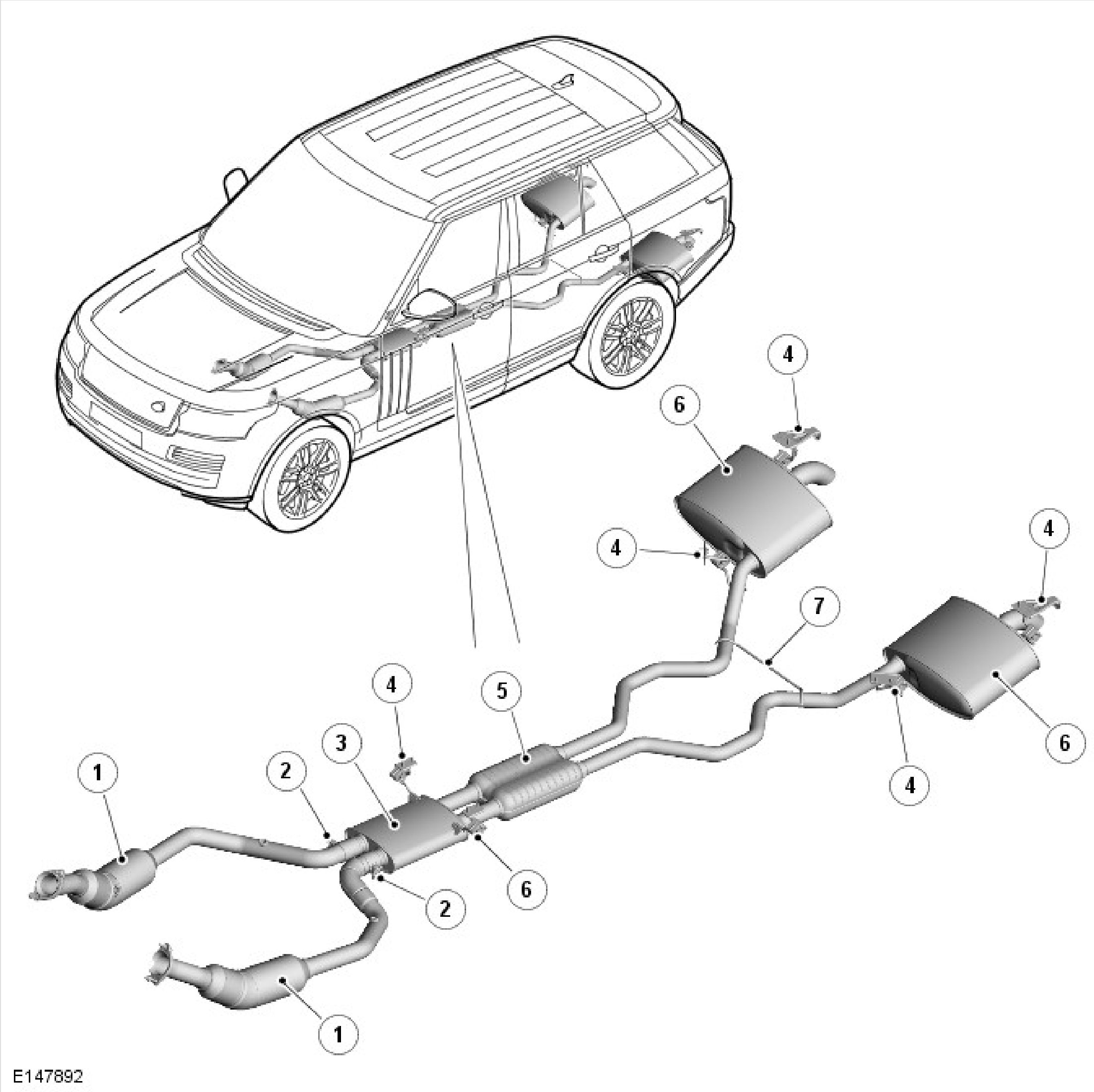
2014.0 RANGE ROVER (LG), 309-00

EXHAUST SYSTEM - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

EXHAUST SYSTEM - 5.0L V8 PETROL N/A (NATURALLY ASPIRATED) AND S/C (SUPERCHARGED)



ITEM	DESCRIPTION
1	Catalytic Converter
2	Clamp
3	Front Silencer
4	Mounting Rubber (6 Off)
5	Centre Resonator Silencer
6	Rear Silencer

7	Exhaust Brace

EXHAUST SYSTEM

The V8 5.0L exhaust system is used on both N/A (Naturally Aspirated) and S/C (Supercharged) vehicles.

The V8 5.0L exhaust system is supplied from the factory as a two-piece assembly and is attached to the underside of the body with six mounting rubbers.

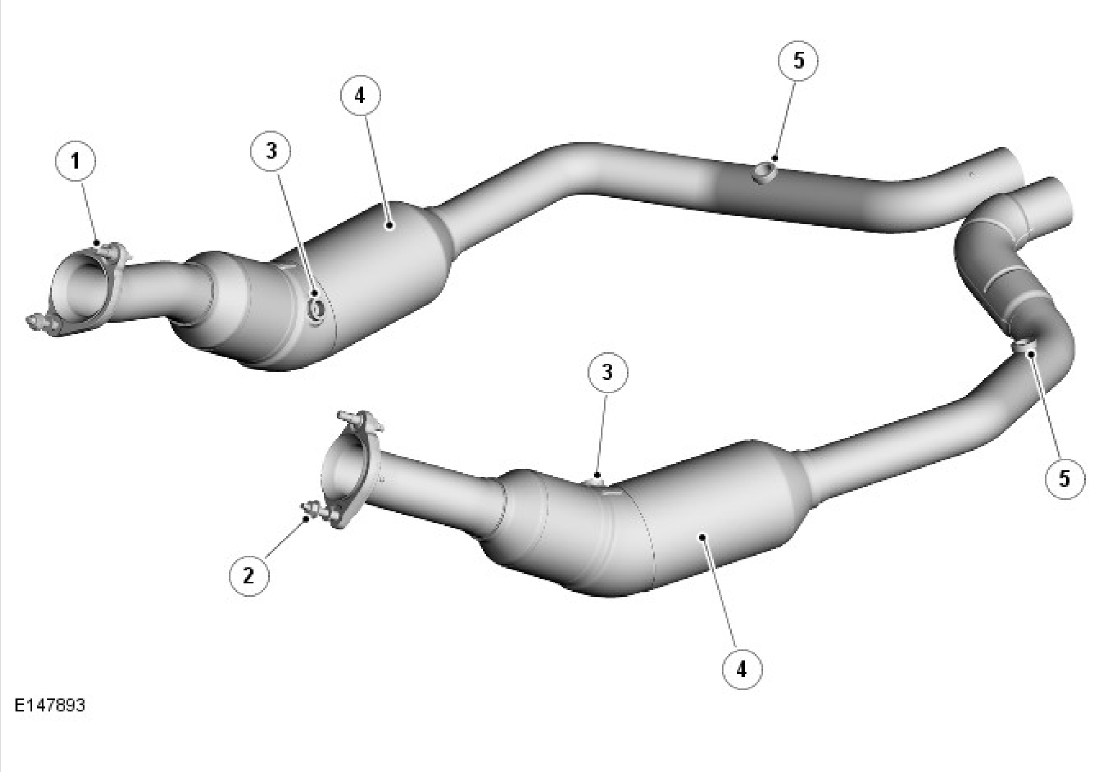
The exhaust front section comprises of two pipes each incorporating a catalytic converter. Each catalytic converter is joined to a inlet pipe with a flange that is attached to each of the exhaust manifold.

The front section is fitted with both pre-catalyst and post-catalyst temperature sensors on each of the pipes.

The two outlet pipes from the front section connect to the inlet pipes of the front silencer and are secured by two clamps. The two pipes from the front silencer connect directly into the centre resonator silencer. The centre resonator silencer houses perforated tubes that are separated by baffle plates and allows the exhaust gasses to exit the resonator silencer via 2 outlet pipes. The two pipes from the centre resonator split the rear system in to two separate rear silencers.

The rear silencers have two perforated tubes which are supported on two baffle plates. The exhaust gasses are expelled from the rear silencer via a single outlet pipe on each of the rear silencers.

CATALYTIC CONVERTER



ITEM	DESCRIPTION
1	Studs (4 Off)
2	Nuts (4 Off)
3	Pre-Catalyst Heated Oxygen Sensor (HO2S) Mounting Boss
4	Catalytic Converter
5	Post-Catalyst HO2S Mounting Boss

The engine management system provides accurately metered quantities of fuel to the combustion chambers to ensure the most efficient use of fuel and to minimise the exhaust emissions. A threaded boss is located on each manifold and a and another threaded boss is located between the starter and the main catalyst to house the pre and post catalyst oxygen sensors. The engine management system monitors the sensors and uses the information to further improve the fuelling and exhaust emissions.

To further reduce the carbon monoxide and hydrocarbons content of the exhaust gases, two catalytic converters are integrated into the front down pipe from each exhaust manifold. In the catalytic converter the exhaust gases are passed through honeycombed ceramic elements coated with a special surface treatment called 'washcoat'. The washcoat increases the surface area of the ceramic elements by a factor of approximately 7000. On top of the 'washcoat' is a coating containing metals, which are the active constituent for converting harmful emissions into inert by-products. The metals add oxygen to the carbon monoxide and the hydrocarbons in the exhaust gases, to convert them into carbon dioxide and water respectively.